

Random Features, Kernel Approximation, and Double Descent on Tabular Regression

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Classical theory. The bias-variance tradeoff predicts that test error rises monotonically once you have more parameters than training points.

Modern practice. Heavily overparameterized models often generalize *better* than smaller ones.

Where does the classical curve break?

- Belkin et al. (2019) showed a *double descent* curve: a spike at the interpolation threshold, then a second descent.
- Schaeffer et al. (2023) attribute the spike to three factors that act jointly.
- This project reproduces the spike on tabular data and tests whether each factor alone removes it.

Three Threads, One Pipeline

The same random-feature ridge pipeline tests three foundational ideas. Each is a separate question we can answer empirically.

Kernel approximation (Rahimi & Recht 2007)

What. An RBF kernel measures similarity by $k(x, x') = \exp(-\|x - x'\|^2/2\sigma^2)$. Powerful but $O(N^2)$ to evaluate at scale. Random Fourier features replace it with P cheap random projections that give an unbiased estimate. **Question.** Does the approximation actually converge to the exact kernel as P grows?

Double descent (Belkin et al. 2019)

What. Test error peaks near the interpolation threshold $P=N$ (where the model just barely fits the training data exactly), then descends *again* as P grows past N . This contradicts the classical bias-variance prediction. **Question.** Can we reproduce the spike on tabular data?

Three-factor explanation (Schaeffer et al. 2023)

What. The spike is hypothesized to arise from three factors: (a) no ridge regularization, (b) test variation concentrated in trailing singular directions of Φ_{train} , (c) noisy targets. Each factor alone is supposedly sufficient to flatten the spike. **Question.** Does each factor really do that on its own?

Cosine variant (Rahimi & Recht 2007). The "cosine" name distinguishes this formulation from the sin/cos paired version of random Fourier features. Each random direction produces *one* cosine feature instead of two paired sin and cos features.

$$\phi(x)_i = \sqrt{\frac{2}{P}} \cos(\omega_i^\top x + b_i), \quad \omega_i \sim \mathcal{N}(0, \sigma^{-2}I), \quad b_i \sim \text{Unif}(0, 2\pi).$$

- **Geometric intuition.** Each ω_i chooses a random direction. The cosine wraps the projection $\omega_i^\top x$ onto the unit circle. Two points close in input space land close on every circle, so their inner product is large.
- Inner product $\langle \phi(x), \phi(x') \rangle$ is an unbiased estimate of $k_\sigma(x, x') = \exp(-\|x - x'\|^2 / 2\sigma^2)$.
- As $P \rightarrow \infty$ the random-feature kernel converges to the exact RBF kernel.
- Bandwidth $\sigma = 2.279$ from the median pairwise distance heuristic on the training subset.

Dataset and Sweep Configuration

Why tabular? Random-feature ridge regression is the cleanest setting for studying double descent. Image and text data introduce architecture-specific confounds (convolutions, attention, etc.) that mask the pure feature/regression dynamics. Tabular regression isolates exactly the behavior the three-factor account predicts.

Dataset. California Housing (UCI, via sklearn). 20 640 rows, 8 standardized features, continuous target.

Setting	Value
Test holdout	20% ($N_{\text{test}} = 4128$)
Training subset N	256
Random feature counts P	$\{4, 8, 16, \dots, 4096\}$ (16 values)
Feature seeds per P	3
Default ridge λ	10^{-3}
Bandwidth σ	2.279 (median heuristic)

$N = 256$ is small on purpose. The interpolation threshold $P = N$ becomes cheap to span with a P grid that crosses it densely.

Methods

Class-method benchmarks

Linear regression, ridge regression on raw features, mean-predictor baseline.

Reference target

Exact RBF kernel ridge regression. The limit that random-feature ridge should approach as $P \rightarrow \infty$.

Main sweep

Random-feature ridge regression at sixteen P values, three feature seeds each.

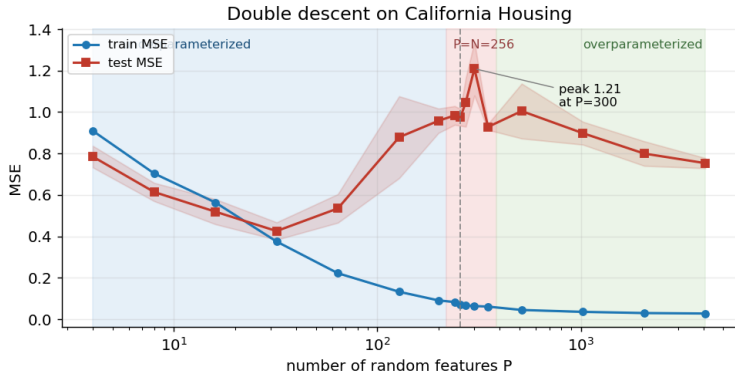
Three-factor ablation (Schaeffer et al. 2023)

(a) ridge regularization at $\lambda \in \{10^{-4}, \dots, 10^{-1}\}$, (b) projection of test features onto the top- k singular directions of Φ_{train} , (c) noiseless target $y_{\text{clean}} = w_{\text{lin}}^{\top} x$ from the best linear fit.

Spectral analysis

SVD of Φ_{train} at every P . Smallest singular value vs. P is the structural witness for factor (a).

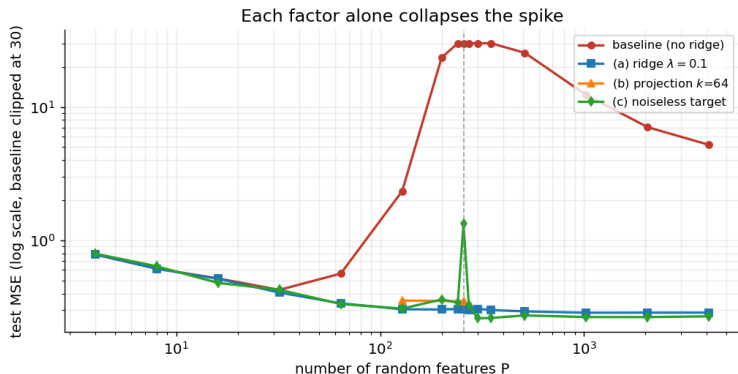
Reproducing Double Descent



x -axis is the number of random features P on a log scale, y -axis is test MSE (lower is better). The shaded bands mark the underparameterized regime ($P \ll N$, blue), the interpolation threshold ($P \approx N$, red), and the overparameterized regime ($P \gg N$, green). The dashed line marks $P = N = 256$.

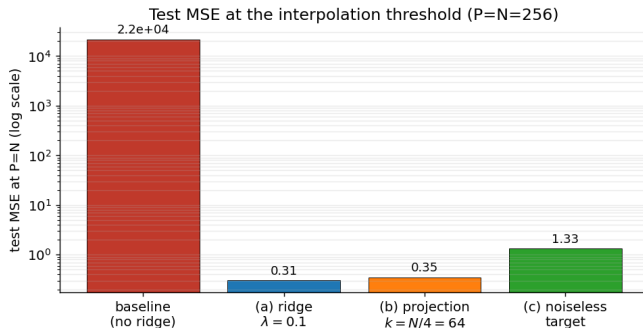
- Best in the underparameterized regime: 0.43 at $P = 32$.
- Spike at the interpolation threshold: 1.21 at $P \approx 300$.
- Second descent in the overparameterized regime: 0.75 at $P = 4096$.

Each Factor Alone Collapses the Spike



All four conditions are plotted on the same log- y axis (lower is better). The **red baseline** curve has the spike at $P = N$. The **blue**, **orange**, and **green** curves are each one of the three Schaeffer factors applied alone. Each factor flattens the spike independently.

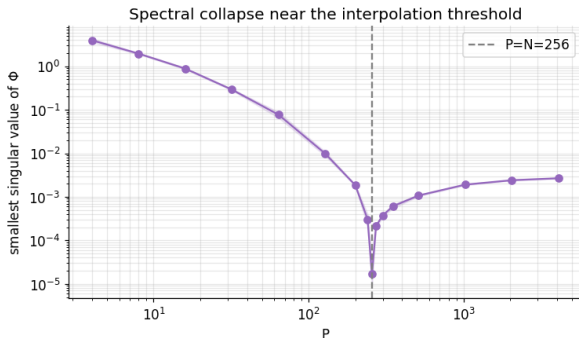
The Headline Numbers at $P = N$



y -axis is test MSE at the interpolation threshold $P = N = 256$, on a log scale because the baseline is four orders of magnitude above the rest (lower is better).

- Baseline (no ridge) blows up to $\sim 2 \times 10^4$ from numerical instability ($\sigma_{\min} \rightarrow 0$).
- Ridge $\lambda = 0.1$ collapses it to 0.31. Four orders of magnitude.
- Projection at $k = N/4$ matches it (0.35) without using λ .
- Noiseless target alone reaches 1.33. Less dramatic but still confirmatory.

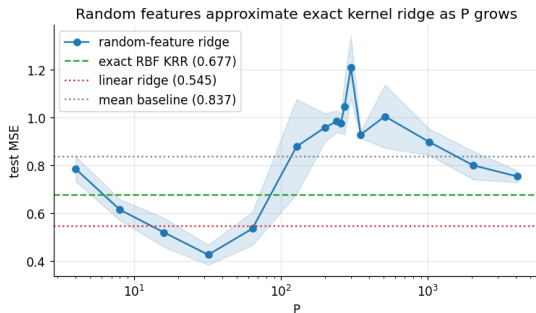
Spectral Collapse at $P = N$



Both axes are log-scale. x -axis is the number of random features P , y -axis is the smallest singular value $\sigma_{\min}$ of the random feature matrix Φ_{train} . The matrix becomes invertible (and ridge regression well-conditioned) only when $\sigma_{\min}$ is comfortably above zero.

- $\sigma_{\min}$: 0.082 at $P = 64$, drops to 6×10^{-7} at $P = N$, recovers to 0.003 at $P = 4096$.
- The dip is exactly at the interpolation threshold.
- Structural witness for factor (a). The small singular value is what ridge

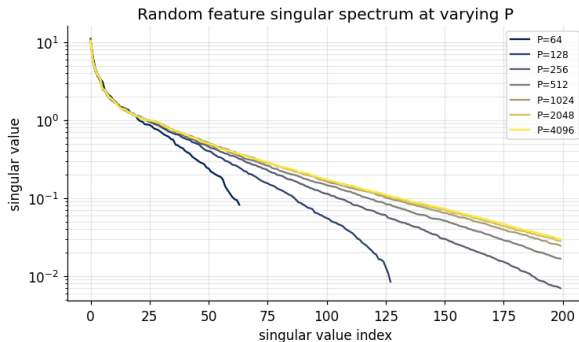
Random Features Approximate Exact RBF Kernel Ridge



The blue curve is the random-feature ridge sweep across P (lower is better). Horizontal lines are fixed-MSE values for the three reference baselines. The curve approaching a horizontal line means random features are approximating that method.

- Random-feature ridge approaches the exact RBF kernel ridge line as P grows. The approximation works.
- Linear ridge wins (0.55). California Housing has strong linear structure and RBF at the median bandwidth is not competitive.
- Approximation point and kernel-choice point are separate. The first holds. The second doesn't.

Singular Spectrum at Varying P



Each colored curve is the full singular spectrum of Φ_{train} at one fixed P . x -axis is the singular value index (largest first), y -axis is the singular value on a log scale. A curve crashing toward the bottom means many near-zero singular values, the geometric source of double descent.

- As P grows the spectrum lengthens. Near $P = N$ the trailing tail collapses to numerical zero.
- In the overparameterized regime the bulk is preserved and the tail recovers.

Putting the Story Together

Going back to the central question. Where does the classical bias-variance curve break?

Answer. It breaks because the ridgeless interpolator at $P = N$ is built on top of a near-singular feature matrix. The spike is a consequence of that singularity, and any of three independent fixes can remove it.

- ❶ Test MSE peaks at $P \approx 300$ (just past $N = 256$) at 1.21, then descends to 0.75 at $P = 4096$.
- ❷ Ridge $\lambda = 0.1$ collapses the spike from $\sim 2 \times 10^4$ to 0.31.
- ❸ Projection at $k = N/4$ reaches 0.35 without using λ .
- ❹ Noiseless target alone reaches 1.33.
- ❺ Smallest singular value of Φ_{train} collapses to 6×10^{-7} exactly at $P = N$. This is the structural witness.

- **Single dataset, single kernel.** California Housing with the RBF kernel only. The picture may differ on data with weaker linear structure or with kernels other than RBF.
- **No bandwidth tuning.** σ is set by the median heuristic only. A tuned bandwidth might close the gap to linear ridge.
- **Three-factor ablations done one at a time.** Factorial combinations are needed to isolate each factor's contribution from the others.
- **Small training subset** ($N = 256$). Chosen to make the interpolation threshold cheap to span, but classical bias-variance behavior may differ at N in the thousands.

- **Synthetic data with controlled noise.** Lets factor (c) be tested cleanly with known signal-to-noise ratio.
- **Tuned bandwidths and additional kernels.** Tests whether the linear-wins-on-California-Housing result is bandwidth-specific or kernel-specific.
- **Larger N grids.** Sweep N to test whether the three-factor account holds across regimes, not just at $N = 256$.
- **Factorial three-factor combinations.** Run all $2^3 = 8$ combinations of (a, b, c) on/off to estimate each factor's marginal contribution.

Five Takeaways

- ① Double descent is reproducible on tabular data with a small training subset. Test MSE peaks at $P \approx 300$ (just past N) at 1.21.
- ② **Ridge regularization** is the most powerful single fix. $\lambda = 0.1$ collapses the spike from $\sim 2 \times 10^4$ to 0.31.
- ③ **Leading-mode projection** at $k = N/4$ matches ridge (0.35) without using λ at all.
- ④ The smallest singular value of Φ_{train} collapses to 6×10^{-7} exactly at $P = N$. The dip is the structural source of the spike.
- ⑤ Random Fourier features genuinely approximate the RBF kernel as P grows. Whether RBF is the right kernel for the data is a separate question.

Questions?

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